

1. O/N/23/22/Q5

Fig. 5.1 shows a device that cools the air in a room by evaporation.

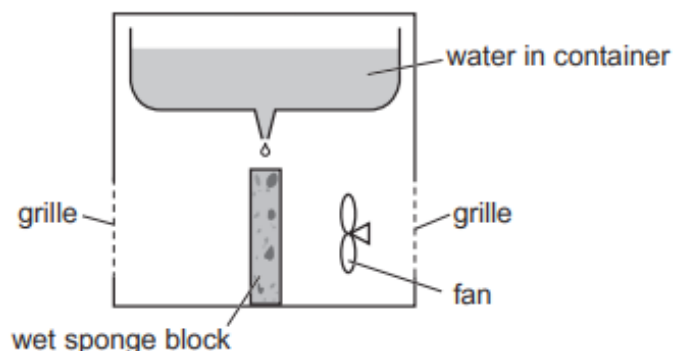


Fig. 5.1

Water drips from the container into the sponge block and keeps it wet.

Air flows in through one grille and out through the other.

As the air flows through the wet sponge block, the water in the sponge block evaporates.

(a) When the water evaporates, the temperature of water remaining in the sponge block decreases.

(i) Explain, in terms of the movement of particles, why this decrease in temperature occurs

.....

.....

.....

..... [2]

(ii) Explain why energy is needed when the water evaporates.

.....

..... [1]

(b) State **two** ways in which evaporation differs from boiling.

1

.....

2

.....

[2]

[Total: 5]

2. M/J/23/22/Q4

Fig. 4.1 shows the particles (atoms) at one instant in a sample of iron at a temperature below its melting point.

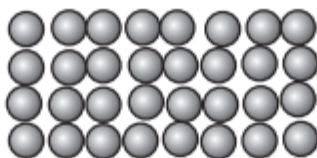


Fig. 4.1

- (a) (i) State the lowest possible temperature on the Celsius scale and on the Kelvin scale.

Celsius scale °C Kelvin scale K [1]

- (ii) The temperature of the solid increases. The sample remains a solid.

Describe how the motion of the particles changes.

.....
 [2]

- (iii) The solid melts.

State what happens to the internal energy and the temperature of the solid as it melts.

internal energy
 temperature [2]

- (b) A student:

- places a 300g piece of iron in boiling water until the iron is at a temperature of 100 °C
- removes the iron from the water and places it immediately into 100 g of water at 25 °C.

The iron cools and the water warms until both reach the same temperature, 44 °C.

The specific heat capacity of water is 4.2 J/(g °C). No energy is lost to the surroundings.

- (i) Calculate the change in energy (internal energy) of the water as it warms up.

change in energy = J [2]

- (ii) Calculate the specific heat capacity of iron.

specific heat capacity = J/(g °C) [2]

[Total: 9]

3. M/J/23/21/Q5

Fig. 5.1 shows the particles (molecules) in a sample of liquid water.

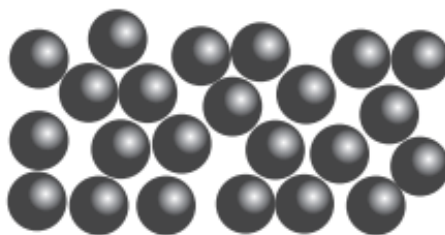


Fig. 5.1

- (a) Explain, using ideas about particles, why liquids expand more than solids for the same temperature rise.

.....
.....
.....
..... [2]

- (b) The boiling point of water is 100°C .

- (i) State the boiling point of water on the Kelvin scale of temperature.

boiling point = K [1]

- (ii) The temperature remains constant as water turns from liquid to gas at the boiling point.

Explain, in terms of particles, why energy must be provided even though the temperature stays constant.

.....
..... [1]

- (c) An electric heater is used to heat a sample of metal, as shown in Fig. 5.2. There is no thermal energy transferred from the metal to the surroundings during the heating.

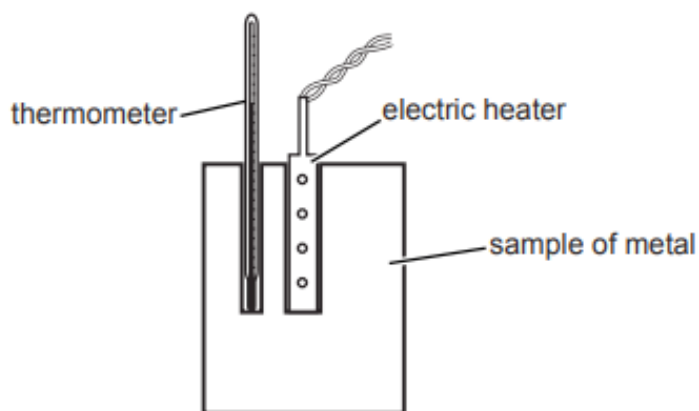


Fig. 5.2

The metal has a mass of 200g and is initially at 15 °C.

The heater is switched on for 2.0 minutes and then switched off. The maximum temperature reached by the metal is 40 °C.

The power of the heater is 20 W.

Calculate the specific heat capacity of the metal.

Show your working.

specific heat capacity = J/(g °C) [3]

[Total: 7]

4. M/J/21/22/Q9

(a) Fig. 9.1 represents the conversion of energy in a mobile phone.



Fig. 9.1

(i) State the form of energy stored in the battery.

..... [1]

(ii) State the useful form of output energy from the battery.

..... [1]

(iii) State **two** useful forms of output energy from the mobile phone.

..... [2]

(b) When the battery is in use, the average current is 1.3 mA.

Determine the charge that passes through the battery in a time of 2.0 minutes.

charge = [2]

- (c) A student estimates the thermal energy produced in the battery when the mobile phone is used and uses this value to estimate the efficiency of the battery.

He measures a temperature rise of 5.0°C within the battery in a period of time when the useful energy output from the battery is 5200 J .

The mass of the battery is 110 g and its specific heat capacity is $830\text{ J/(kg }^{\circ}\text{C)}$.

- (i) Calculate the thermal energy (heat) needed to raise the temperature of the battery by 5.0°C .

thermal energy = [3]

- (ii) State what is meant by *efficiency*.

.....
..... [2]

- (iii) Calculate the efficiency of the battery.

efficiency = [2]

- (iv) Suggest and explain **one** reason why the value for the efficiency calculated in (iii) is larger than the actual efficiency of the battery.

.....
.....
.....
..... [2]

[Total: 15]

5. M/J/21/21/Q3

A hot steel rod is cooled by plunging it into cold water, as shown in Fig. 3.1.

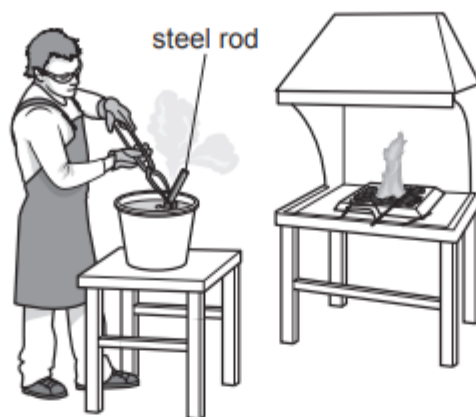


Fig. 3.1

- (a) The steel rod has a mass of 2.0 kg and is initially at a temperature of 500°C . It cools to 50°C when placed in the water.

The specific heat capacity of steel is $460\text{ J/(kg }^\circ\text{C)}$.

Calculate the thermal energy (heat) lost by the steel rod as it cools to 50°C .

thermal energy = [3]

- (b) A small mass of water boils when the rod is placed in the water. The remaining water then cools to room temperature and some of it evaporates.

In both boiling and evaporation, water molecules escape into the air.

- (i) State **one** way in which boiling is *different* from evaporation.

.....
.....
..... [1]

- (ii) The rate of evaporation decreases as the water cools.

Explain why this happens, using ideas about molecules.

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.....
..... [2]

6. O/N/20/21/Q9

A small glass measuring cylinder of oil is placed inside a freezer where the temperature is -18°C . Fig. 9.1 shows how the temperature of the oil varies with time t .

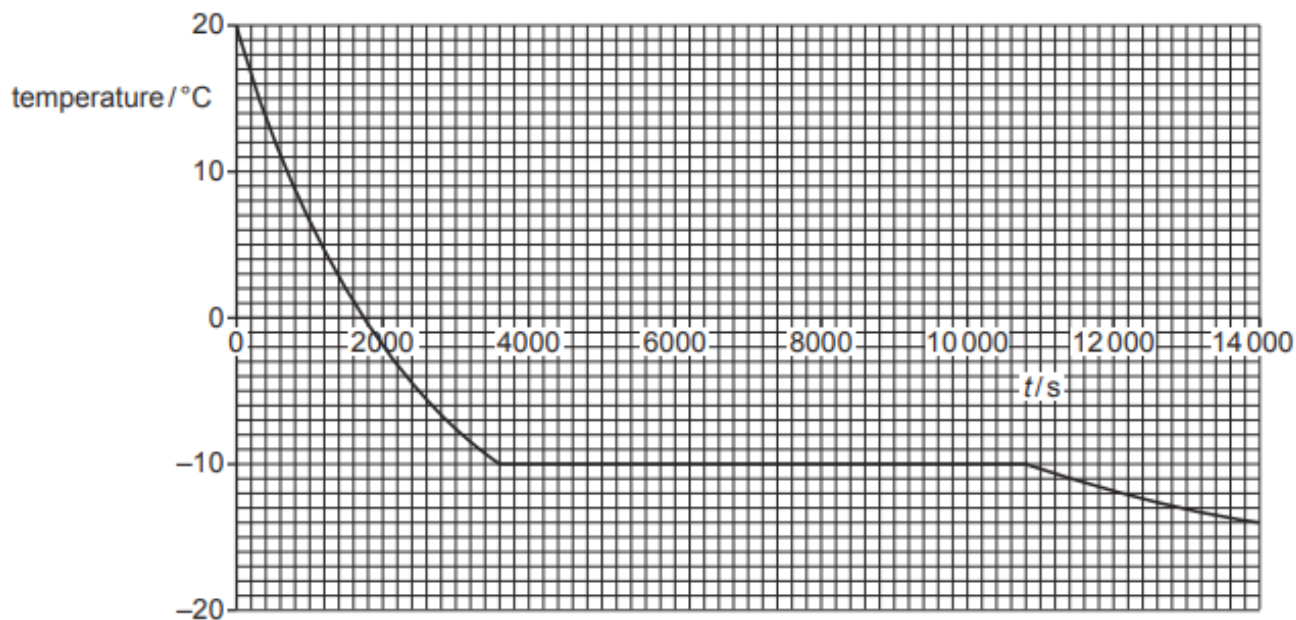


Fig. 9.1

Fig. 9.1 shows that it takes 700s for the temperature to decrease from 20°C to 10°C but that it takes 1900s to decrease from 0°C to -10°C .

(a) Suggest why these times are different.

.....

.....

.....

..... [3]

(b) Explain what happens to the molecules of the oil and what happens to the level of the oil in the glass measuring cylinder as the temperature decreases from 20°C to 0°C .

.....

.....

.....

..... [3]

- (c) (i) Determine the melting point of the oil.

..... [1]

- (ii) Explain, in terms of molecules, why the temperature of the oil does not change between $t = 3600\text{ s}$ and $t = 10800\text{ s}$.

.....
.....
.....
..... [3]

- (d) There is 45g of oil in the glass measuring cylinder and the specific latent heat of fusion (melting) of the oil is $5.7 \times 10^4\text{ J/kg}$.

Calculate:

- (i) the energy transferred from the oil between $t = 3600\text{ s}$ and $t = 10800\text{ s}$

energy = [2]

- (ii) the average rate at which energy is transferred from the oil between $t = 3600\text{ s}$ and $t = 10800\text{ s}$.

rate = [1]

- (e) The graph in Fig. 9.1 is steeper before the horizontal section than it is after.

Use this observation to compare the specific heat capacity of oil in the liquid and solid states and explain your reasoning.

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.....
.....
..... [2]

[Total: 15]

7. O/N/20/22/Q4

(a) Describe, in terms of molecules, the structure of a solid.

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..... [3]

(b) Fig. 4.1 shows a metal-worker heating a horseshoe to a high temperature before shaping it.

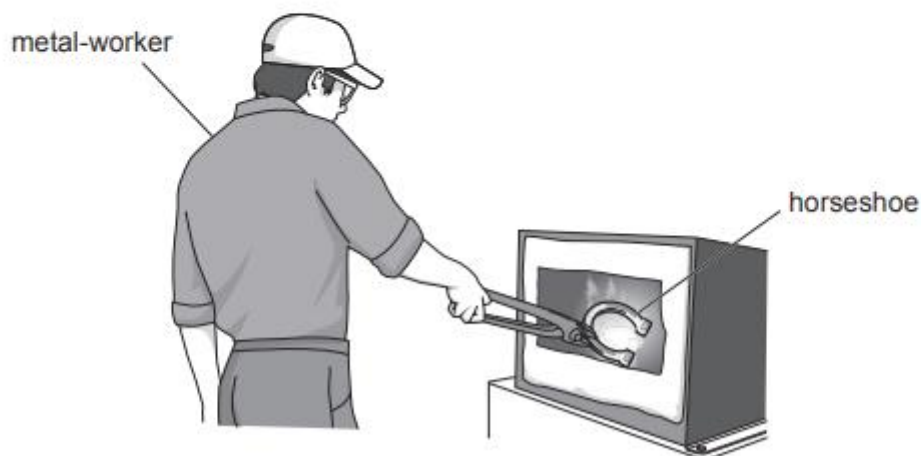


Fig. 4.1

The metal-worker then cools the horseshoe by dropping it into 8.0 kg of water at 18 °C. The final temperature of the water is 43 °C.

The specific heat capacity of water is 4200 J/(kg °C).

(i) Calculate the thermal energy transferred to the water as the horseshoe cools.

energy = [2]

(ii) The heat capacity of the horseshoe is 850 J/°C.

Calculate the temperature of the horseshoe immediately before it is dropped into the water.

temperature = [3]

8. O/N/19/21/Q3

A saucepan of water is placed on an electric hotplate and the water is brought to the boil.

As the water boils, it changes from a liquid into a gas.

(a) Describe, in terms of the molecules, **three** ways in which a gas differs from a liquid.

1.
.....
2.
.....
3.
.....

[3]

(b) In order to keep the liquid boiling, it is necessary to keep the hotplate switched on.

Explain why energy must be supplied in order to turn a liquid at its boiling point into a gas.

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.....
.....

[2]

[Total: 5]

9. O/N/19/22/Q4

In many countries, solid salt is produced by trapping sea-water in large, shallow ponds and letting the water evaporate.

Fig. 4.1 shows salt being produced in this way.



Fig. 4.1

(a) State two ways in which evaporation differs from boiling.

1.
.....

2.
.....

[2]

(b) Describe, in terms of water molecules, what is happening as the water evaporates.

.....
.....
..... [2]

(c) (i) State why the ponds used in this process have large surface areas.

.....
..... [1]

(ii) State why this method of salt production does not work well in a country with a cold climate.

.....
..... [1]

[Total: 6]

10. M/J/18/21/Q10

(a) (i) Describe two differences between boiling and evaporation.

1.

.....

.....

2.

.....

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[4]

(ii) Explain, using ideas about molecules, why thermal energy is needed to boil a liquid.

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.....

.....

.....[2]

(b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16°C

temperature of water after heating = 37°C

volume of water collected in 1.0 minute = $4.6 \times 10^{-3} \text{ m}^3$

specific heat capacity of water = $4200 \text{ J/(kg }^{\circ}\text{C)}$

density of water = 1000 kg/m^3

(i) Calculate the mass of water leaving the shower in 1.0 s.

mass =[2]

- (ii) Calculate the thermal energy (heat) gained by the water in 1.0 s.

energy =[3]

11. O/N/17/21/Q3

Unwanted gold jewellery is heated in a furnace until it melts. Fig. 3.1 shows the liquid gold being poured into a mould to make a gold bar.

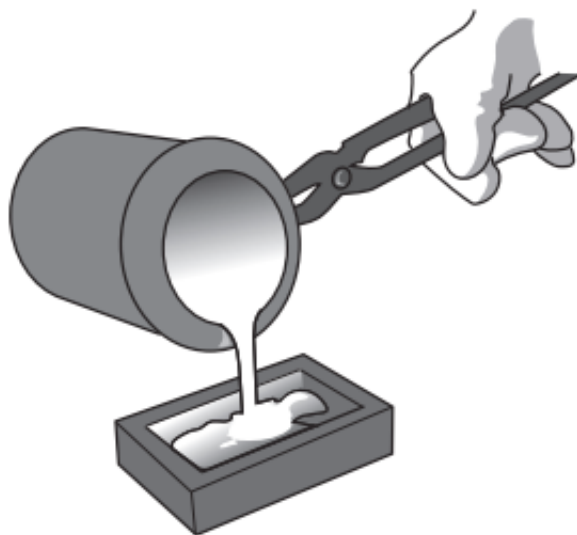


Fig. 3.1

The melting point of gold is 1100°C .

- (a)** State two ways in which the molecular structure of gold at 1200°C differs from its structure at 1000°C .

1.

.....

2.

.....

[2]

- (b)** In the furnace, 0.84 kg of solid gold reaches its melting point.

The furnace continues to supply thermal energy and, after a short time, all the gold melts.

- (i)** Explain, in terms of the molecules, why thermal energy is needed to melt the gold.

.....

.....

.....

..... [2]

- (ii)** The specific latent heat of fusion of gold is 64 kJ/kg.

Calculate the thermal energy supplied to melt all the gold.

thermal energy = [2]

12. O/N/17/22/Q10

A boiler containing water is plugged into a 230 V mains supply. The heater in the boiler is operating at full power.

(a) The power used by the heater is 2.8 kW.

(i) Calculate the current in the heater.

current =[3]

(ii) The wiring in the cable from the socket and in the boiler remains safe and does not overheat provided that the current is less than 20 A.

Suggest a suitable fuse rating for the circuit.

.....[1]

(b) The specific heat capacity of water is $4200 \text{ J/(kg } ^\circ\text{C)}$. The heater is used to increase the temperature of 6.3 kg of water.

(i) In a certain time, the temperature of the water increases from 23°C to 49°C .

Calculate the change in the internal energy of the water.

change in internal energy =[2]

(ii) The heater supplies thermal energy to the water at a constant rate. It takes more time for the temperature of the water to increase from 49°C to 75°C than from 23°C to 49°C . In both cases, the increase in temperature is the same.

Suggest why it takes more time for the temperature to increase from 49°C to 75°C .

.....
.....
.....[1]

(c) Fig. 10.1 shows the heater at the bottom of the water in the boiler.

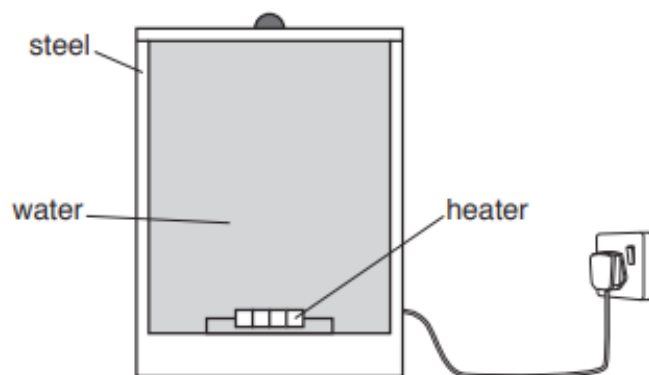


Fig. 10.1

(i) Explain how thermal energy is transferred from the heater throughout the water.

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.....[3]

(ii) The part of the boiler that holds the water is made from steel.

Describe, in terms of particles, how thermal energy passes through the steel.

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.....[3]

(iii) State and explain, in terms of particles, what happens to the density of the steel as its temperature increases.

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.....[2]

13. M/J/17/22/Q4

Fig. 4.1 shows apparatus used to measure the specific heat capacity of a metal.

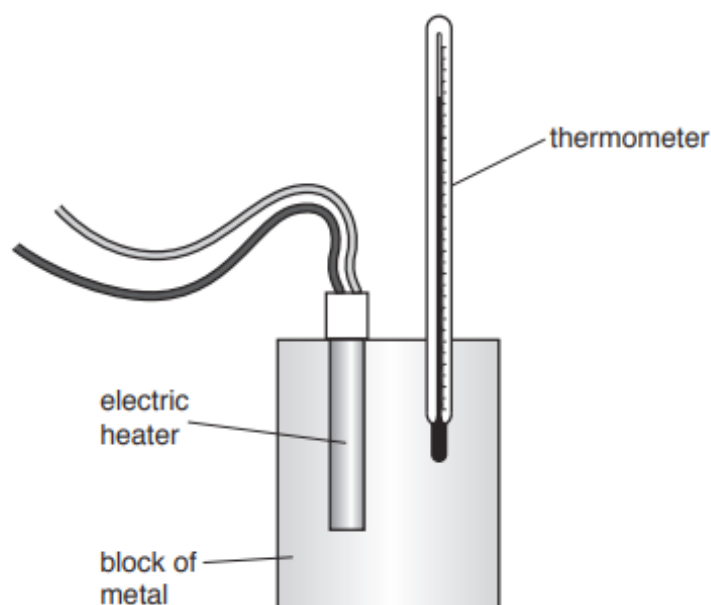


Fig. 4.1

The electric heater is switched on and supplies 17 000 J of thermal energy (heat) to the block of metal. The temperature of the metal rises from 16 °C to 38 °C. The mass of the block of metal is 850 g. No energy is lost from the metal.

- (a) Calculate the specific heat capacity of the metal.

specific heat capacity =[2]

- (b) Calculate the heat capacity of the block of metal.

heat capacity =[1]

- (c) Explain how thermal energy (heat) is conducted through the metal.

.....
.....
.....[1]

14. O/N/16/21/Q10

A particular type of cooking oil is a liquid at room temperature.

- (a) At room temperature, air is easily compressed but the oil cannot be compressed.

Explain, in terms of molecules,

- (i) why the density of the oil is much greater than that of a gas,

.....
.....
.....[1]

- (ii) why the oil cannot be compressed.

.....
.....
.....[1]

- (b) Some of the oil is poured into a metal pan. The oil in the pan has a mass of 680 g.

- (i) The density of the oil is 0.85 g/cm^3 .

Calculate the volume of oil in the pan.

volume =[2]

- (ii) The pan is placed on an electric hotplate, as shown in Fig. 10.1.

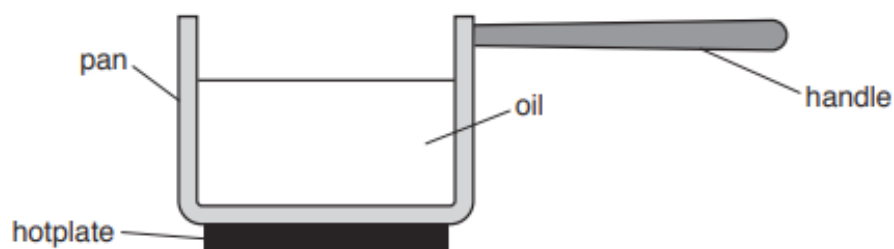


Fig. 10.1

1. Explain, in terms of molecules or free electrons, how thermal energy is conducted through the metal base of the pan.

.....

.....

.....

.....

.....

.....[3]

2. Explain how thermal energy spreads to all the oil in the pan.

.....

.....

.....

.....

.....

.....[3]

3. Suggest and explain a suitable material for the handle of the pan.

.....

.....

.....[1]

- (c) The specific heat capacity of the oil is $2.0 \text{ J/(g } ^\circ\text{C)}$. The oil is heated from 20°C to its boiling point of 260°C .

(i) State what is meant by *boiling point*.

.....
.....[1]

(ii) Calculate the thermal energy needed to heat the 680 g of oil to its boiling point.

thermal energy =[2]

(iii) Suggest one reason why the energy output from the hotplate, when heating the oil, is larger than the value calculated in (c)(ii).

.....
.....
.....[1]

15. O/N/16/22/Q4

Fig. 4.1 shows a kettle containing water placed on the burner of a gas cooker.



Fig. 4.1

The gas burner is lit at time $t = 0$. At $t = 250\text{ s}$ the temperature of the water is 100°C , the boiling point of water.

(a) State what is meant by *boiling point*.

.....
.....[1]

(b) Fig. 4.2 shows how the temperature of the water changes with time t .

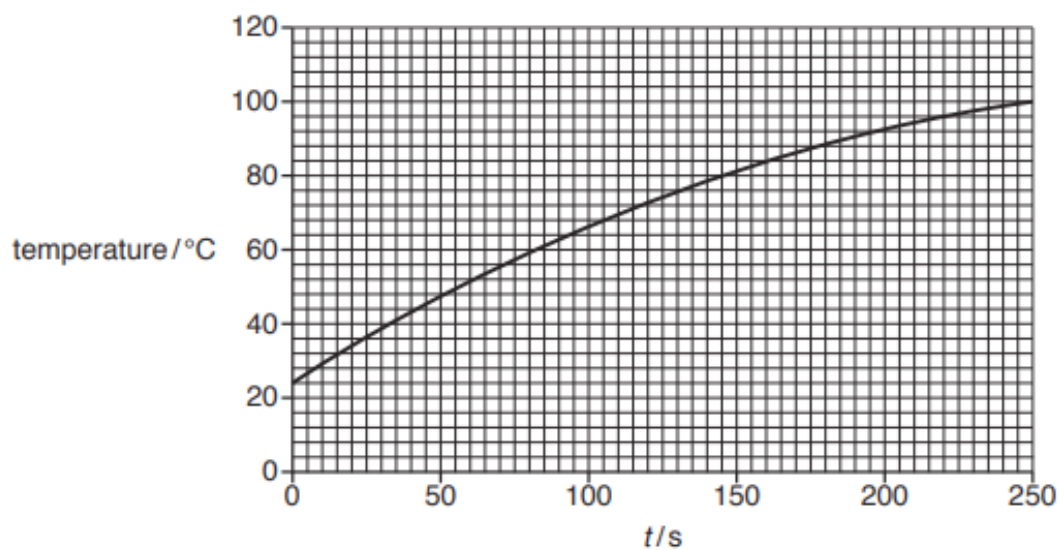


Fig. 4.2

- (i) The kettle contains 1.5 kg of water which has a specific heat capacity of $4200 \text{ J/(kg } ^\circ\text{C)}$.

Using Fig. 4.2, determine the increase in the internal energy of the water between $t = 0$ and $t = 250 \text{ s}$.

increase in internal energy =[3]

- (ii) Thermal energy (heat) is transferred to the water at a constant rate but the temperature of the water increases at a rate that is not constant, as shown in Fig. 4.2.

Explain why the temperature increases in this way.

.....
.....
.....
.....
.....[2]

- (c) When the temperature reaches 100°C , the kettle is left on the burner. Thermal energy is still supplied to the water. The water boils as the molecules form bubbles and rise to the surface.

- (i) State what happens to the temperature of the water.

.....
.....[1]

- (ii) Explain, in terms of the molecules, why it is necessary to supply thermal energy in order to keep the water boiling.

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.....
.....
.....
.....[2]

16. M/J/16/21/Q10

(a) (i) State what is meant by *melting point*.

.....
.....[2]

(ii) The temperature of a liquid is between its melting point and its boiling point.

Using ideas about molecules, describe the changes that occur when the liquid evaporates.

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.....
.....
.....[3]

(b) The boiling point of nitrogen is -196°C .

Liquid nitrogen, below its boiling point, is stored in a vacuum flask, as shown in Fig. 10.1.

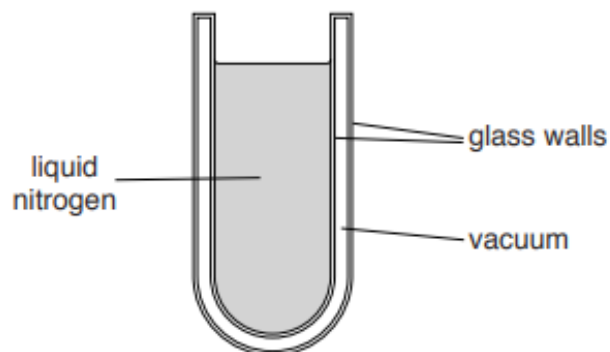


Fig. 10.1

The flask has two glass walls with a vacuum between them.

(i) Suggest where, in the apparatus shown in Fig. 10.1, evaporation occurs.

.....[1]

(ii) State and explain the advantage of having a vacuum between the two glass walls.

.....

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.....

.....

.....[3]

17. O/N/15/21/Q5

In a steam cleaner, water is heated to its boiling point and is then turned into steam (water in the gas state).

(a) Define *boiling point*.

.....
..... [1]

(b) (i) Describe the molecular structure of a liquid.

.....
.....
.....
.....
..... [2]

(ii) Explain, using ideas about molecules, why thermal energy (heat) must be supplied to the water as it boils.

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.....
.....
.....
..... [2]

18. M/J/15/22/Q4

A metal can and a plastic bottle, both containing liquid, are cooled by placing them in a jug of melting ice, as shown in Fig. 4.1.



Fig. 4.1

The can and bottle each contain 330 g of the same liquid at 15 °C.

- (a) The specific heat capacity of the liquid is 4.2 J/(g °C).

Calculate the thermal energy released when 330 g of the liquid at 15 °C cools to 2 °C.

energy = [2]

- (b) When water at 0 °C is used in the jug, instead of the melting ice, the cooling is slower.

Explain why cooling is faster when using melting ice in the jug, rather than water at 0 °C.

.....
.....
.....
..... [2]

- (c) The liquid in the metal can cools down faster than the liquid in the plastic bottle.

Suggest why this happens.

.....
.....
..... [1]

19. M/J/15/21/Q10

(a) A student makes a diagram of the molecules in a liquid, as shown in Fig. 10.1.

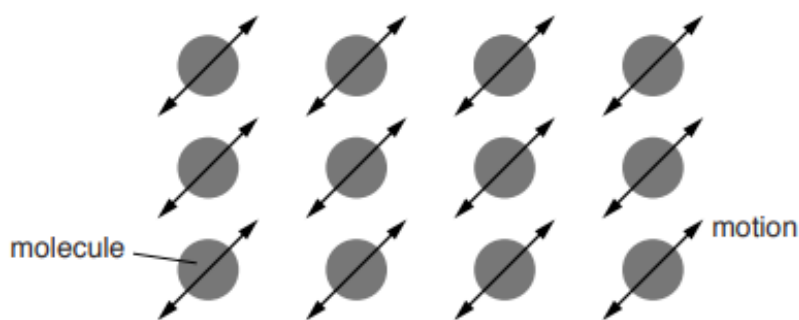


Fig. 10.1

Suggest two ways in which the student's diagram does not show the positions and motion of the molecules accurately.

1.
.....
2.
.....[2]

(b) Some of the molecules of a liquid evaporate.

(i) Describe what happens during evaporation.

.....
.....[1]

(ii) Explain why evaporation has a cooling effect on the liquid.

.....
.....
.....
.....[2]

- (c) The beaker shown in Fig. 10.2 contains pure water at a temperature of 80°C . A thermometer, initially at room temperature, is placed in the air just above the water. The reading on the thermometer starts to increase slowly.

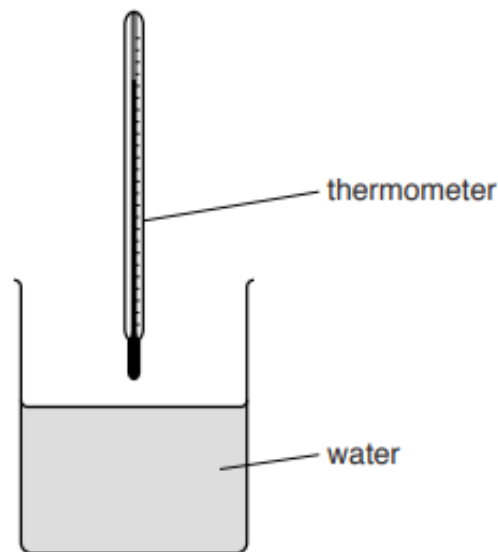


Fig. 10.2

- (i) Explain how convection in the air causes the reading on the thermometer to increase.

.....
.....[1]

- (ii) A heater is placed underneath the beaker and the water boils vigorously to produce steam. The thermometer reading increases quickly.

Explain how the water in the gas state gives thermal energy (heat) to the thermometer.

.....
.....
.....
.....[2]

20. O/N/14/22/Q6

- (a) State what happens to the molecules of a gas as its temperature increases.

.....
.....[1]

- (b) Fig. 6.1 shows a runner in a long distance race. He tips water over himself to keep cool.



Fig. 6.1

- (i) Explain, in terms of molecules, how the evaporation of water keeps the runner cool.

.....
.....
.....
.....[3]

- (ii) At one point in the race, a strong breeze blows past the runner and the water evaporates more quickly.

Suggest one reason why the water evaporates more quickly in the breeze.

.....
.....[1]

21. M/J/14/22/Q3

Most substances expand when they are heated.

- (a) (i)** State one example where expansion is useful.

..... [1]

- (ii)** State one example where expansion causes a problem.

..... [1]

- (b)** Explain, using ideas about molecules, why solids expand when heated.

.....
.....
.....
.....
..... [2]

- (c)** When equal volumes are heated through the same temperature rise, the expansions of solids, liquids and gases are different.

Complete each of the two sentences using one of these expressions:

much larger slightly larger much smaller slightly smaller

1. The expansion of a solid is than the expansion of a liquid.

2. The expansion of a gas is than the expansion of a liquid.
[2]

22. M/J/14/21/Q2

Fig. 2.1 shows apparatus used when determining the specific heat capacity of aluminium.

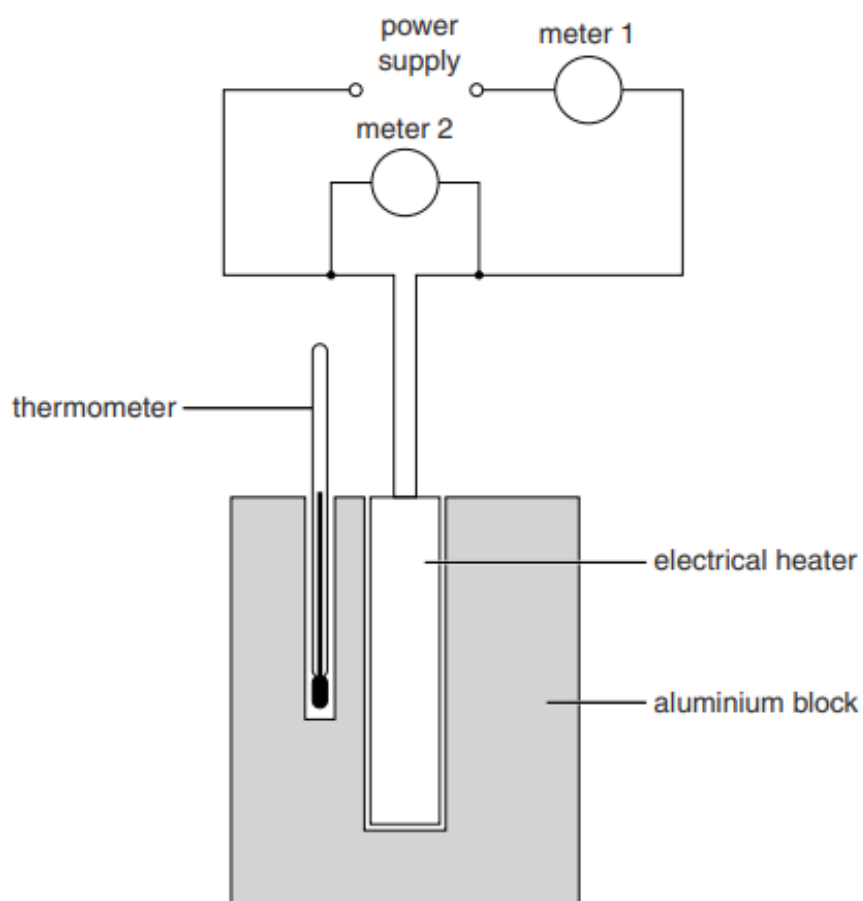


Fig. 2.1

Meter 1 and meter 2 are electrical meters.

(a) State the quantities measured by meter 1 and by meter 2.

meter 1:

meter 2:

[1]

(b) The following results are obtained in the experiment:

initial temperature of block	= 18 °C
final temperature of block	= 40 °C
mass of block	= 0.85 kg
energy supplied to heater	= 17 000 J

Assuming that no heat is lost from the block during the experiment, calculate the specific heat capacity of aluminium.

specific heat capacity = [2]

(c) In practice, some heat is lost from the block and the air above it becomes hot.

(i) Describe and explain the process by which the hot air moves away from the block.

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.....
.....
..... [2]

(ii) Suggest how to reduce the loss of heat from the block in the experiment.

.....
..... [1]

23. O/N/13/21/Q1

A swimming pool is filled with 450m^3 of water at 12°C .

The density of water is 1000kg/m^3 and its specific heat capacity is $4.2\text{J/(g}^\circ\text{C)}$.

(a) Calculate the mass of water in the swimming pool.

mass =[2]

(b) (i) Calculate the thermal energy (heat) needed to raise the temperature of the water to 27°C .

thermal energy =[3]

(ii) Suggest one reason why the thermal energy supplied to the water by the heating system, when raising the temperature of the water to 27°C , differs from the value calculated in **(i)**.

.....
.....[1]

24. M/J/13/22/Q11

A student places a small electrical heater inside a cup of water, as shown in Fig. 11.1.

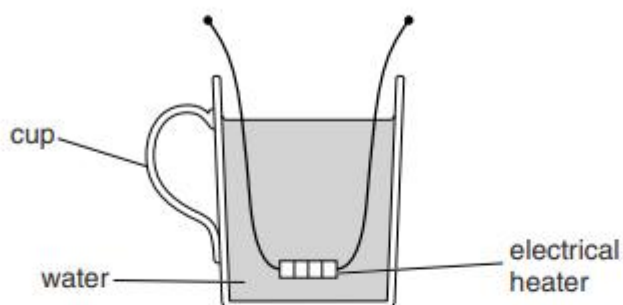


Fig. 11.1

The student determines the electrical power of the heater.

- (a) In the space above the cup on Fig. 11.1, draw the electrical circuit that the student uses. Include an ammeter, a voltmeter and a power supply. [2]
- (b) The voltage of the power supply is 12V and the current is 4.2A.
- (i) Calculate the electrical power input to the heater.

power = [2]

- (ii) Calculate the energy input to the heater in 8.0 minutes. Give your answer in kWh.

energy = kWh [3]

(c) During heating, the student notices that some of the water evaporates from the cup.

(i) Describe, using ideas about molecules of water, what happens during evaporation.

.....
.....
..... [2]

(ii) The student finds that the amount of evaporation increases when the temperature of the water is higher.

State and explain one other change that increases the amount of evaporation.

.....
.....
..... [2]

(iii) State two differences between evaporation and boiling.

1.
.....
2.
.....
..... [2]

(d) The student turns off the power supply and the water cools.

Describe and explain how convection in the air causes the water to cool.

.....
.....
.....
..... [2]

25. M/J/12/22/Q4

A beaker contains 60g of a hot substance, initially in the liquid state. Fig. 4.1 shows how the temperature of the substance changes with time t as it cools in a laboratory.

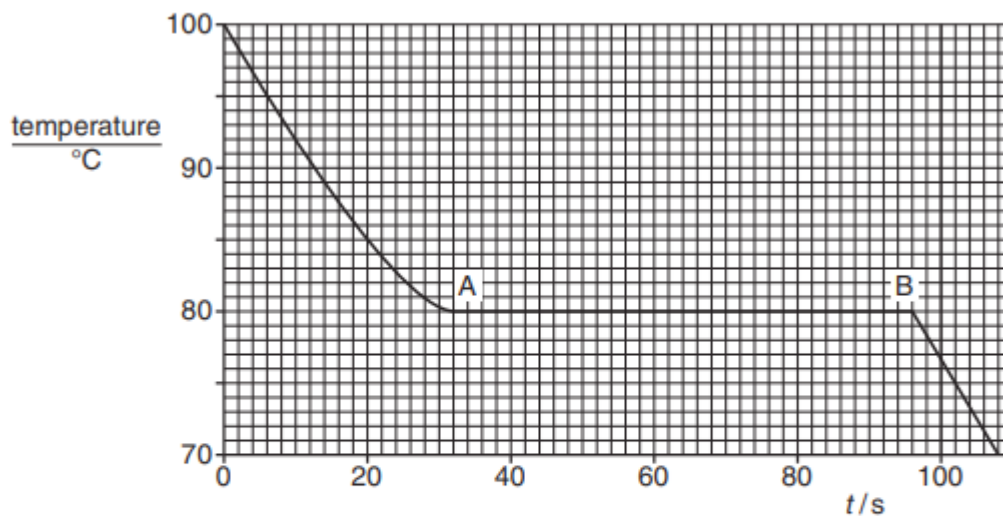


Fig. 4.1

- (a) Use Fig. 4.1 to determine the melting point of the substance.

.....[1]

- (b) The specific heat capacity of the liquid is $1.7 \text{ J/(g}^{\circ}\text{C)}$. Calculate the loss of thermal energy (heat) from the liquid between $t = 0$ and $t = 20 \text{ s}$.

loss of thermal energy =[2]

- (c) Between points A and B on Fig. 4.1, the temperature is constant as the substance changes from liquid to solid.

- (i) Explain why the temperature stays constant, even though thermal energy is lost by the substance.

.....
[2]

- (ii) Describe the change in the arrangement of the molecules as the substance changes from a liquid to a solid.

.....

 [2]

26. M/J/12/21/Q4

A student slowly heats a sample of solid wax in a test-tube.

Fig. 4.1 shows how the temperature of the wax varies with time t .

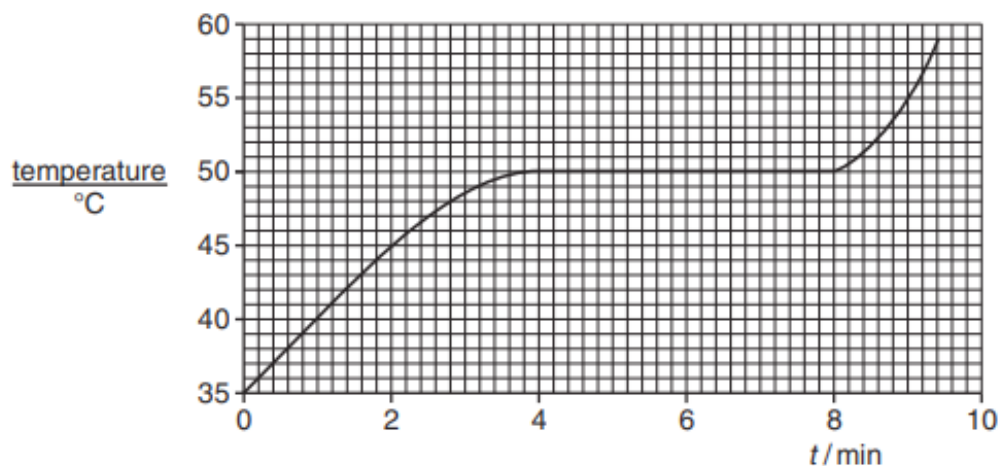


Fig. 4.1

- (a) State what is meant by the *melting point* of a substance.

.....
[1]

- (b) Thermal energy (heat) passes into the wax throughout the experiment.

- (i) Describe what effect this energy has on the wax between $t = 0$ and $t = 4$ min.

.....
[1]

- (ii) Describe and explain what effect this energy has on the wax between $t = 4$ min and $t = 8$ min.

.....

[2]

- (c) The wax expands as the temperature rises. It expands more between $t = 8$ min and 9 min than between $t = 0$ and 1 min. Explain why.

.....
[1]

27. M/J/11/22/Q3

(a) Explain, using ideas about molecules,

(i) why a balloon filled with gas expands when heated,

.....
.....
.....
.....[2]

(ii) why a balloon filled with water expands very little when heated.

.....
.....[1]

(b) (i) A bubble of gas rises from the bottom of a lake to the surface. The pressure at the bottom of the lake is $3.0 \times 10^5 \text{ Pa}$ and the pressure at the surface is $1.0 \times 10^5 \text{ Pa}$. The volume of the bubble at the bottom of the lake is 2.0 cm^3 .

Calculate the volume of the bubble at the surface.

volume =[2]

(ii) State one assumption that you have made in your calculation in (i).

.....
.....[1]

28. M/J/11/22/Q11

- (a) In an experiment to measure the specific heat capacity of water, an electric heater heats water in a glass beaker. The temperature of the water is measured at regular intervals of time. Fig. 11.1 shows how the temperature varies with time t .

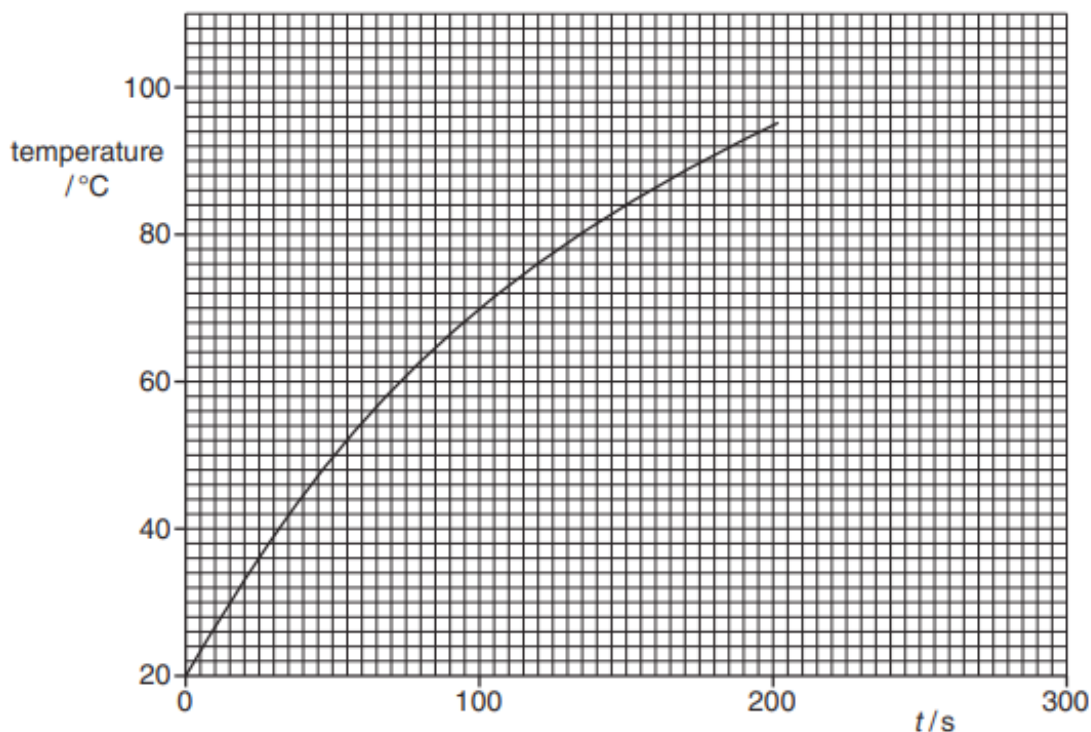


Fig. 11.1

- (i) Use Fig. 11.1 to determine the change in temperature between

$t = 0$ and $t = 100$ s,

change =

$t = 100$ s and $t = 200$ s.

change =

[1]

- (ii) State and explain why the values in (i) are different.

.....

 [2]

- (iii) Describe and explain what happens to the water if the heating is continued.

.....

 [2]

- (b) (i)** The experiment in **(a)** is repeated using 72g of water. The heater supplies 7400J of thermal energy (heat) to the water and the temperature rise of the water is 23°C.

Calculate the specific heat capacity of water.

specific heat capacity =[2]

- (ii)** A bullet of mass 72g is fired from a gun at a speed of 450 m/s.

Calculate the kinetic energy of the bullet.

energy =[3]

- (iii)** The amount of internal energy gained by the water and the amount of kinetic energy gained by the bullet are approximately equal.

Describe the change in the motion of the molecules of the water and of the molecules of the bullet that this addition of energy has caused.

water:

.....

bullet:

.....

[3]

29. M/J/10/22/Q4

A student notices puddles of water on a road, as shown in Fig. 4.1.



Fig. 4.1

Later in the day, he passes the puddles again and some of the water has evaporated.

(a) State two changes to the atmospheric conditions that would cause the water to evaporate faster.

1.

2.

[2]

(b) Explain, in terms of molecules, what happens during evaporation.

.....

.....

.....

.....

..... [2]

30. O/N/08/Q3

A river flows over a cliff, producing a waterfall. The water, at a temperature of 7.2°C , falls 700m into a pool. The gravitational field strength is 10 N/kg .

- (a) Calculate the change in the gravitational potential energy of each kilogram of water due to its fall.

energy change = [2]

- (b) Assume that all of this energy is converted into thermal energy (heat) in the water in the pool. Calculate the temperature of this water when it is in the pool. The specific heat capacity of water is $4200\text{ J/(kg }^{\circ}\text{C)}$.

temperature = [3]

31. O/N/07/Q5

Fig. 5.1 shows the arrangement of atoms in a solid block.

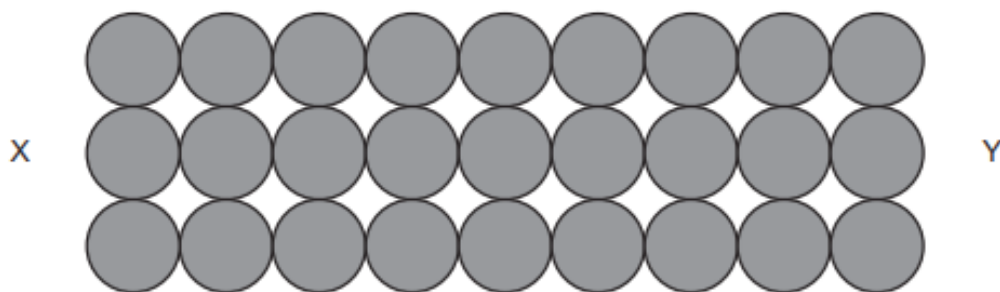


Fig. 5.1

(a) End X of the block is heated. Energy is conducted to end Y, which becomes warm.

(i) Explain how heat is conducted from X to Y by the atoms.

.....
.....
..... [2]

(ii) Explain why the solid block expands when it is heated.

.....
.....
..... [1]

(b) The block is heated and becomes a liquid.

Describe the changes that occur to the arrangement and the motion of the atoms.

.....
.....
.....
..... [2]

32. M/J/04/Q3

Thermal expansion can be a problem.



gap in
bridge

Fig. 3.1

Small gaps are left in railway lines and in bridges, as shown in Fig. 3.1.

(a) Explain in detail why the gaps are needed.

.....

.....

.....

.....

.....[3]

(b) State one other problem caused by thermal expansion and explain how it can be solved.

.....

.....

.....[1]